Dot Product

Section 11.3

8/19/26

A) Mechanics

1) Dot Product

$\vec{U} = \langle u_1, u_2, u_3 \rangle$

$\vec{V} = \langle v_1, v_2, v_3 \rangle$

$$\boxed{u \cdot v = U_1 \cdot V_1 + U_2 \cdot V_2 + U_3 \cdot V_3 = \#}$$

$u \cdot v = 0 \Rightarrow u \perp v$

$v \cdot v = \|v\|^2$

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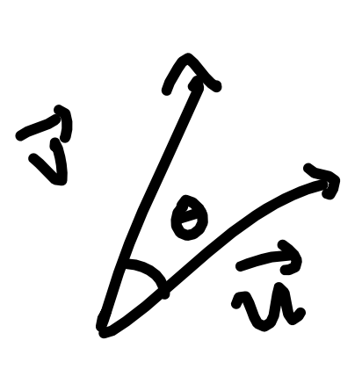
2) angle between vectors



$$\Rightarrow \boxed{\cos\theta = \frac{u \cdot v}{\|u\| \cdot \|v\|}}$$

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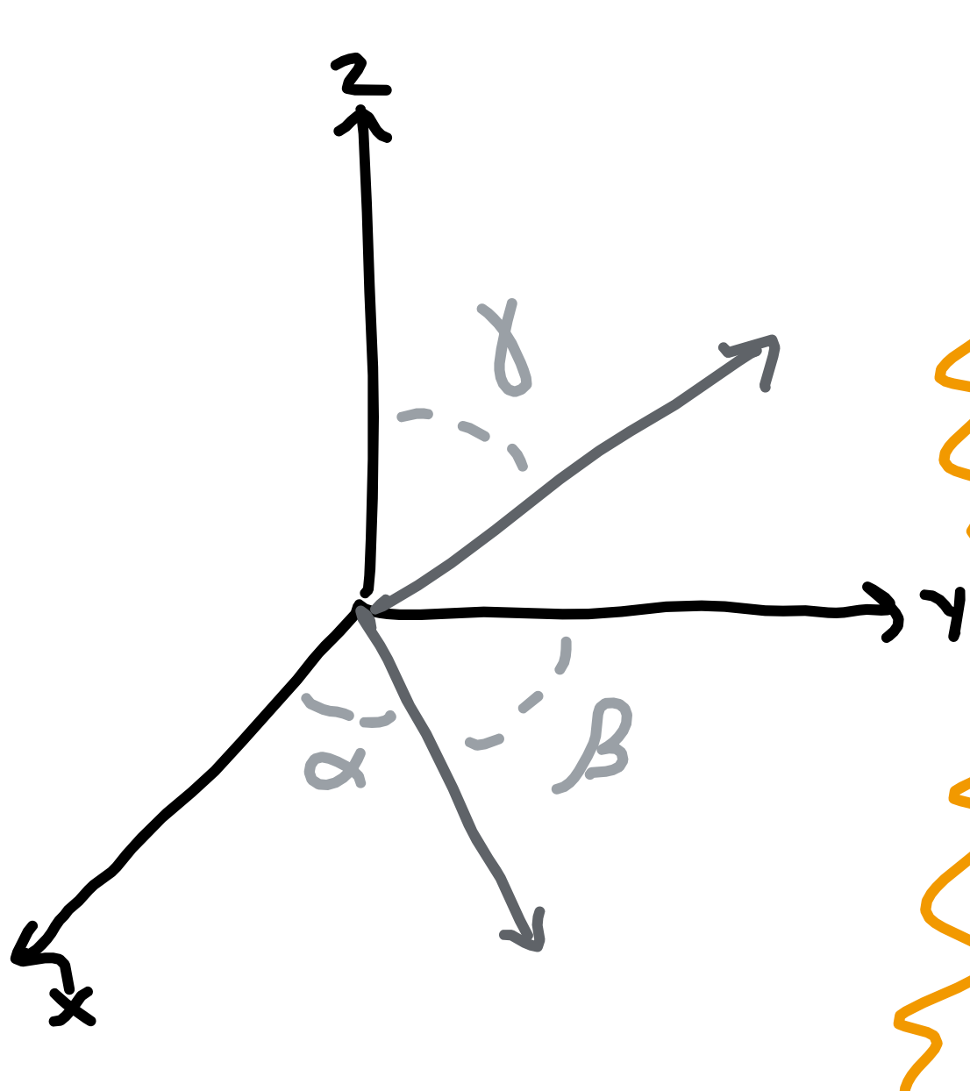
3) angle between axis



α ∡ to x-axis

$\cos\alpha = \frac{v_1}{\|v\|}$

β ∡ to y-axis

$\cos\beta = \frac{v_2}{\|v\|}$

γ ∡ to z-axis

$\cos\gamma = \frac{v_3}{\|v\|}$

Direction cosines

$\left\langle \frac{v_1}{\|v\|}, \frac{v_2}{\|v\|}, \frac{v_3}{\|v\|} \right\rangle \rightarrow$ Unit Vector

$\langle \cos\alpha, \cos\beta, \cos\gamma \rangle \rightarrow$ Direction Cosine Vector

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ex) Find direction ∡s

$\vec{V} = \langle -4, 3, 5 \rangle \quad \|v\| = \sqrt{50}$

$\left\langle \frac{-4}{\sqrt{50}}, \frac{3}{\sqrt{50}}, \frac{5}{\sqrt{50}} \right\rangle$

$\alpha = \arccos\left(\frac{-4}{\sqrt{50}}\right) \quad \beta = \arccos\left(\frac{3}{\sqrt{50}}\right) \quad \gamma = \arccos\left(\frac{1}{\sqrt{2}}\right)$

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5) a) $(2 \cdot 0) + (-3 \cdot 6) + (4 \cdot 5) = -2$

b) $\left(\sqrt{4+9+16}\right)^2 = 29$

c) 29

d) $\cdot 2 \langle 0, 6, 5 \rangle = \langle 0, -12, 10 \rangle$

e) $(2 \cdot 0) + (-3 \cdot 12) + (4 \cdot 10) = -4$

13) a) $\cos\theta = \frac{2}{\sqrt{18}}$

17) $U \cdot V = \cos\left(\frac{\pi}{3}\right) \cdot 8 \cdot 5 = 40\cos\left(\frac{\pi}{3}\right)$

21) $u = \langle 0, 1, 6 \rangle$ not multiples

$v = \langle 1, -2, -1 \rangle \quad U \cdot V \neq 0$ Neither

23) $u = \langle 2, -3, 1 \rangle$ not multiples

$v = \langle -1, -1, -1 \rangle \quad U \cdot V = 0 \Rightarrow u \perp v$ Orthogonal

27) $A = \langle 2, 0, 1 \rangle \quad B = \langle 0, 1, 2 \rangle \quad C = \langle -0.5, 1.5, 0 \rangle$

$AB = \langle -2, 1, 1 \rangle \quad \vec{AB} \cdot \vec{BC} = 1 + 0.5 - 2 = -0.5$

$BC = \langle -0.5, 0.5, -2 \rangle$

obtuse

$CA = \langle 2.5, -1.5, 1 \rangle$

29) $u = \langle 1, 2, 2 \rangle$

$\cos\alpha = \frac{1}{\sqrt{9}} = \frac{1}{3}$

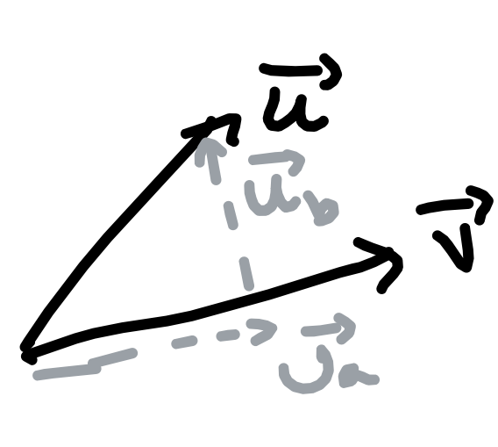
$\cos\beta = \frac{2}{\sqrt{9}} = \frac{2}{3}$

$\cos\gamma = \frac{2}{\sqrt{9}} = \frac{2}{3}$

$\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = 1$

B) Projection Vectors

Project $\vec{u}$ onto $\vec{v}$



$\vec{u} = \vec{u}_a + \vec{u}_b$

$\vec{u}_a = \|\vec{u}\| \cdot \cos\theta \cdot \frac{\vec{v}}{\|v\|} = \cancel{\|\vec{u}\|} \cdot \frac{u \cdot v}{\cancel{\|u\|} \cdot \|v\|} \cdot \frac{\vec{v}}{\|v\|}$

$\vec{u}_a = \frac{u \cdot v}{\|v\|^2} \cdot \vec{v}$ | $\vec{u}_b = \vec{u} - \vec{u}_a$

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ex) project $\vec{b}$ onto $\vec{a}$

$b = \langle 1, 1, 2 \rangle \quad a = \langle -2, 3, 1 \rangle$

$\vec{b}_a = \frac{a \cdot b}{\|a\|^2} \cdot \vec{a} = \frac{3}{14} \langle -2, 3, 1 \rangle$

find component of $\vec{b}$ orthogonal to $\vec{a}$

$b = \vec{b}_a + \vec{b}_b$

$\langle 1, 1, 2 \rangle = \left\langle \frac{-6}{14}, \frac{9}{14}, \frac{3}{14} \right\rangle + \langle x, y, z \rangle$

$\vec{b}_b = \left\langle \frac{20}{14}, \frac{5}{14}, \frac{25}{14} \right\rangle$

C) Application: Work

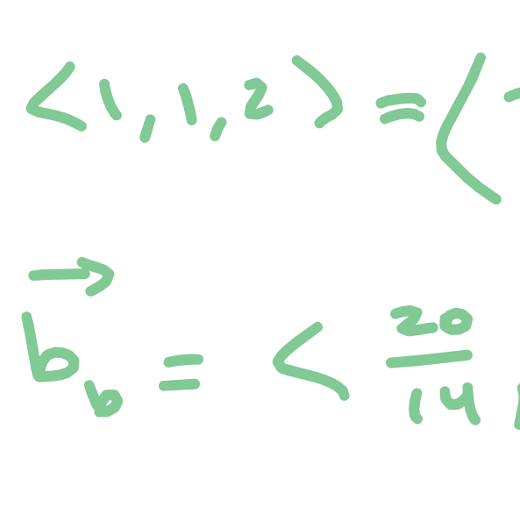
$W = F \cdot d$

What if F was a vector?

a displacement vector

move from P to Q

$\vec{PQ} = \vec{D}$



$w = \left(\overset{F}{\|F\| \cos\theta}\right) \cdot \left(\overset{D}{\|PQ\|}\right)$

$= \|F\| \cos\theta \|PQ\|$

$= \vec{F} \cdot \vec{PQ}$

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ex) Force given by $F = \langle 3, 4, 5 \rangle$ moves from $P(2, 1, 0)$ to $Q(4, 6, 2)$. Find work done

$\vec{PQ} = \langle 2, 5, 2 \rangle$

$W = \langle 3, 4, 5 \rangle \cdot \langle 2, 5, 2 \rangle = 36$

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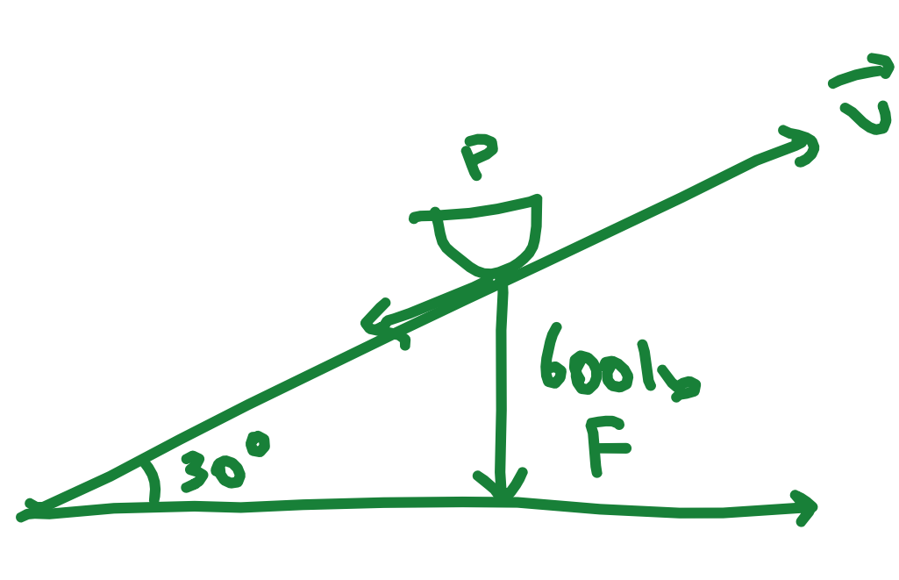
ex) 600lb boat, ramp of 30°

What force is needed to keep boat from rolling?



$V = \langle \cos 30°, \sin 30° \rangle$

$= \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle$

Project F onto v

$\vec{F} = \langle 0, -600 \rangle$

$\frac{F \cdot V}{\|V\|^2} \|\vec{V}\| = \frac{F \cdot V}{\|V\|} = F \cdot V = 300$

(abs. value)

$\|V\| = 1$

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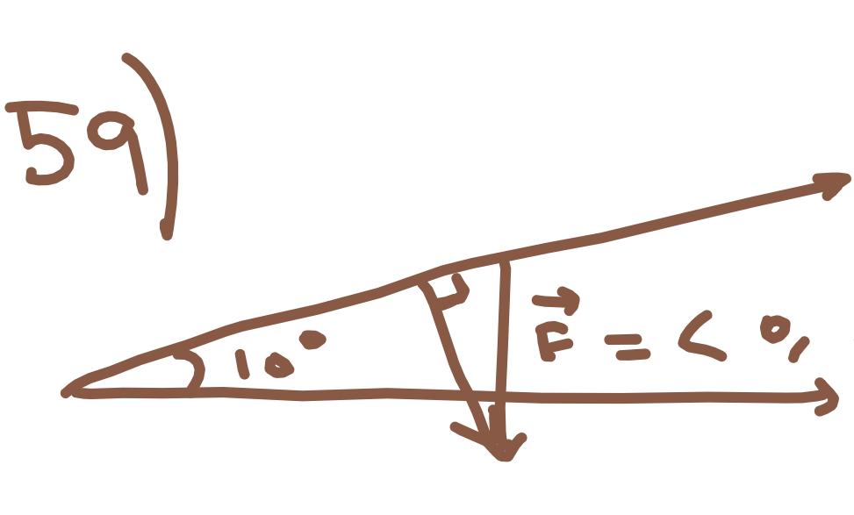
59)



$\vec{v} = \langle \cos 10°, \sin 10° \rangle$

$\vec{F} = \langle 0, -48000 \rangle$

a) $\vec{F}_v = \frac{F \cdot V}{\|v^2\|} \cdot \vec{V}$

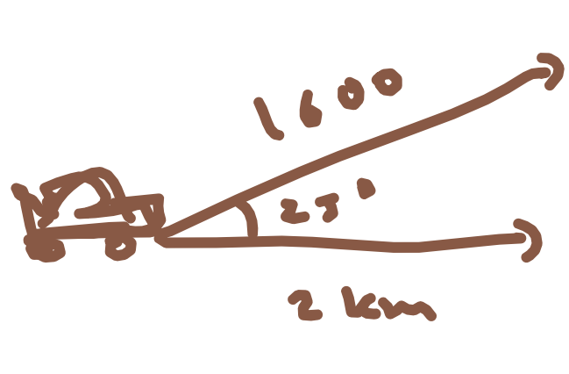
$\|F_v\| = 48000 \sin 10°$

b) $\|\vec{F} - \vec{F}_v\| = 48000^2 - (48000 \sin 10°)^2$

$= \sqrt{48000^2 (1 - \sin^2 10°)} = \sqrt{48000^2 (\cos^2 10°}$

$= \boxed{48000 \cos 10°}$

63)



$\vec{V} = 1600 \langle \cos 25°, \sin 25° \rangle$

$\vec{D} = \langle 2000, 0 \rangle$

$\vec{V} \cdot \vec{D} = 1600 \cdot 2000 \cdot \cos 25°$